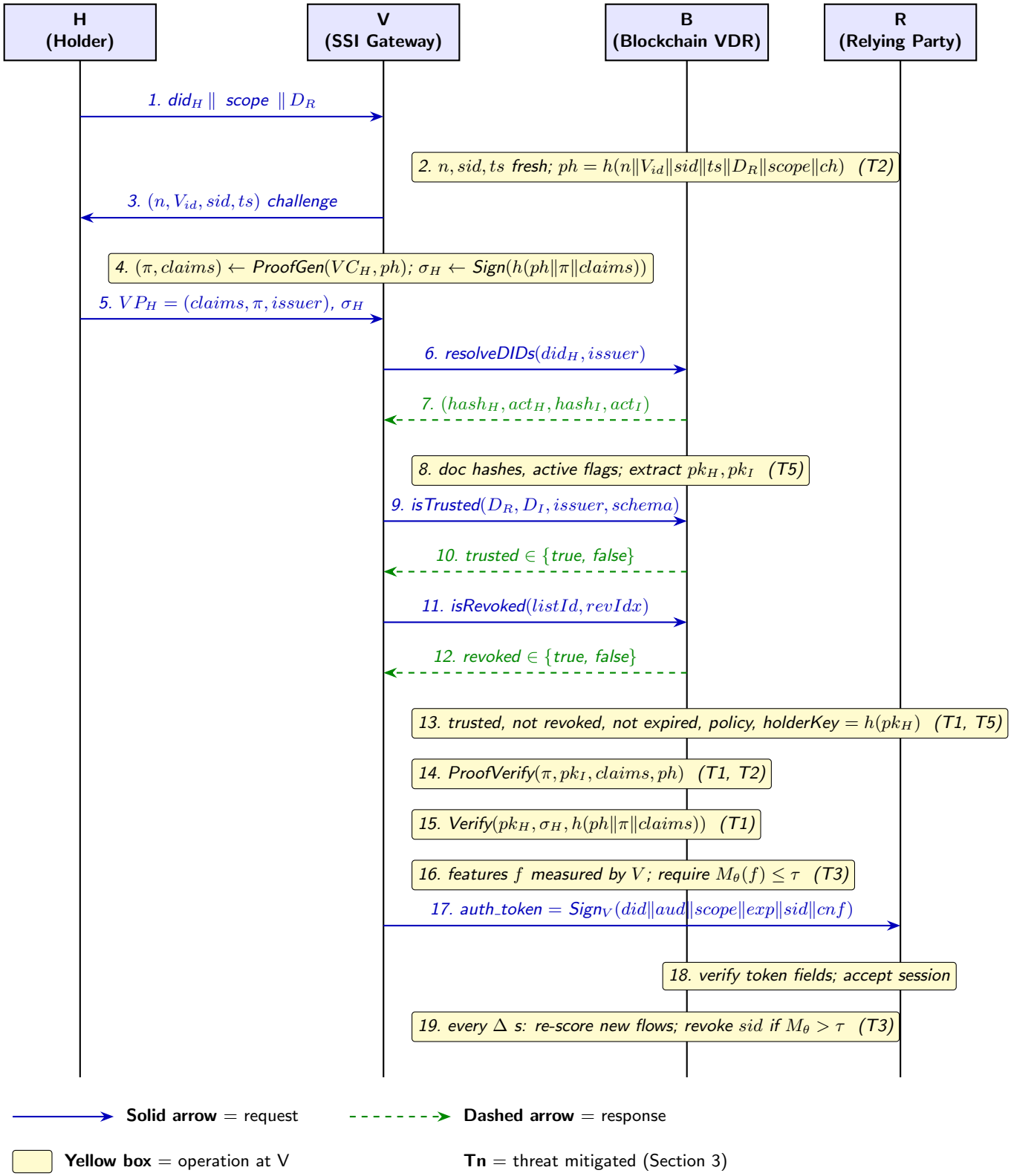

H
(Holder)
V
(SSI Gateway)
B
(Blockchain VDR)
R
(Relying Party)
1. $did_H \parallel$ scope $\parallel D_R$
2. $n, sid, ts$ fresh; $ph = h(n \| V_{id} \| sid \| ts \| D_R \| scope \| ch)$ (T2)
3. $(n, V_{id}, sid, ts)$ challenge
4. $(\pi, claims) \leftarrow ProofGen(VC_H, ph)$; $\sigma_H \leftarrow Sign(h(ph \| \pi \| claims))$
5. $VP_H = (claims, \pi, issuer)$, $\sigma_H$
6. $resolveDIDs(did_H, issuer)$
7. $(hash_H, act_H, hash_I, act_I)$
8. doc hashes, active flags; extract $pk_H, pk_I$ (T5)
9. $isTrusted(D_R, D_I, issuer, schema)$
10. trusted $\in$ {true, false}
11. $isRevoked(listId, revIdx)$
12. revoked $\in$ {true, false}
13. trusted, not revoked, not expired, policy, holderKey $= h(pk_H)$ (T1, T5)
14. $ProofVerify(\pi, pk_I, claims, ph)$ (T1, T2)
15. $Verify(pk_H, \sigma_H, h(ph \| \pi \| claims))$ (T1)
16. features $f$ measured by $V$; require $M_\theta(f) \leq \tau$ (T3)
17. auth_token $= Sign_V(did \| aud \| scope \| exp \| sid \| cnf)$
18. verify token fields; accept session
19. every $\Delta$ s: re-score new flows; revoke $sid$ if $M_\theta > \tau$ (T3)
Solid arrow = request
Dashed arrow = response
Yellow box = operation at V
Tn = threat mitigated (Section 3)